Modelling Postgres Performance Degradation

on Burstable Cloud Instances

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June 18, 2026

The Appeal of Burstable Instances

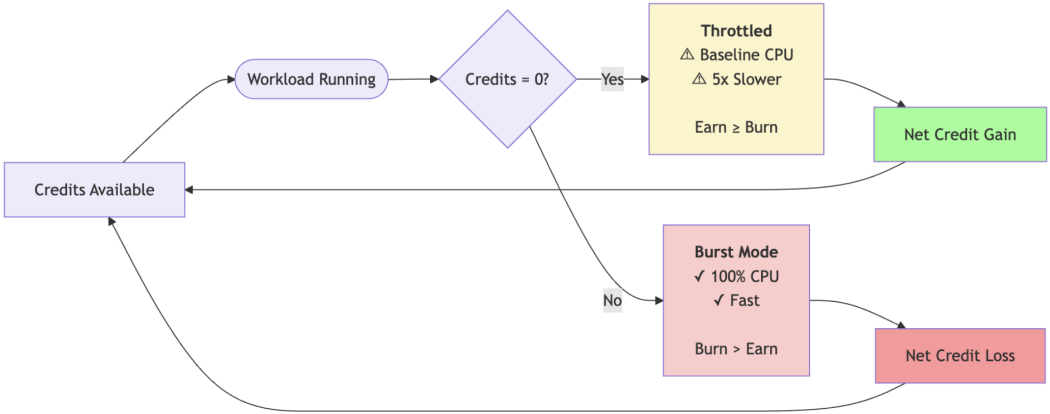
Cost-Effective Cloud Computing

- Azure B-series (Burstable PostgreSQL)
- Similar offerings: AWS T-series, GCP E2 (burstable)
- 20–60% cheaper than fixed-performance instances

The Core Question

Perfect for variable workloads. . . right?

Token Bucket Algorithm



The CPU Credit Model

How it works:

1. Earn credits continuously over time
2. Burn credits when using CPU above baseline
3. Burst to 100% CPU when credits available
4. Throttle to baseline (10–30%) when depleted

Example Blueprint: Azure B2ms

- Base CPU Performance of VM: 30%
- Earns 36 credits/hour, burns 2 credits/min at burst

Azure B-series V1 CPU Burst

Size Name	Base CPU Performance of VM (%)[1]	Initial Credits (Qty.)	Credits banked/hour (Qty.)	Max Banked Credits (Qty.)
Standard_B1ls	5%	30	3	72
Standard_B1s	10%	30	6	144
Standard_B1ms	20%	30	12	288
Standard_B2s	20%	60	24	576
Standard_B2ms	30%	60	36	864
Standard_B4ms	22.5%	120	54	1,296
Standard_B8ms	17%	240	81	1994
Standard_B12ms	17%	360	121	2,908
Standard_B16ms	17%	480	162	3,888
Standard_B20ms	17%	600	202	4,867

Reference Documentation:

The Hidden Danger

Not a crash. Not an error. Just. . . slowness.

When CPU credits deplete:

- Database throttles to baseline CPU;
- Query time increases significantly (5x minimum, often worse);
- Postgres does not know it's being throttled;
- Slow queries hold connections longer.

Result

Connection pool exhaustion.

The Cascading Failure

Watch how one problem triggers the next:

When connections held longer:

- Connection pool saturates;
- New requests queue;
- P99 latency spikes and app layer times out;
- max_connections limit hit;
- PostgreSQL rejects connections.

Result

At this point, you are experiencing a severe production outage.

The Restart Loop

But wait - it gets worse:

When PostgreSQL rejects connection:

- Health checks fail;
- Orchestrator on Azure restarts the database;
- Boot sequence burns remaining CPU credits;
- Comes back up already slow and immediately saturates again;
- Manual intervention required to break the loop.

The Challenge: Incomplete Information

You can calculate credit rates. You can monitor credit balance. But predicting interactions requires testing.

Over-provision	**Load test all configs**	**Simulate**
Skip the B-series entirely and pay for General Purpose.	Trial by fire. Test every possible configuration.	Use the math to model the traffic. Test 50 scenarios in hours.
✓ Safe	✓ Accurate	✓ Good enough
× Expensive at scale	× Weeks of work	✓ Fast

Simulation Approach

Models credit mechanics, pooling modes, and queueing behavior

Simulation is a filter, not a total replacement for reality.

- Reduces load testing scope (test 3 finalists, not 50 configs)

Simulation finds candidates. Load testing validates the finalists.

Why Discrete Event Simulation (DES)?

Definition

DES: Models systems by following each event step by step.

DES models pooling, CPU credits, queueing in burstable instances.

Modelling as Discrete Event Simulation

Burstable instances are not magic. They follow a strict mathematical model called the Token Bucket Algorithm.

Generator	Queue	Server
Load Generator	DB Connection Pool	Burstable Instance

What is SNA?

A lightweight open-source library for Discrete Event Simulations in C# and .NET 10.

Available to model Azure Database for PostgreSQL B-series with M/M/c/K:

- Deterministic logic for stochastic inputs;
- Configurable baseline/earn rates (adapts to any B-series specs);
- Validated against queueing theory predictions.

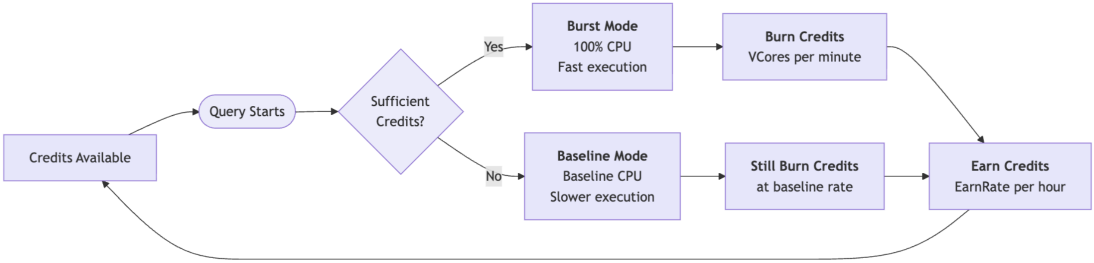
Project repo:

CPU Credits Simulation

Question:

Where is the point where specific workload transitions from "Bursting" to "Death Spiral"?

Credit Mechanics

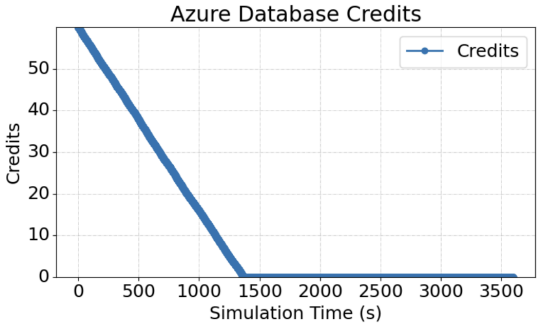


Credit Depletion Timeline

What we're seeing:

- Credits start at 60;
- Linear depletion as burst usage exceeds earn rate;
- Credits hit zero at 1,300 seconds;
- **This is your burst tolerance window: ∼20 minutes.**

**Note: Starting credits affect timeline. Tune this parameter to match your environment.*



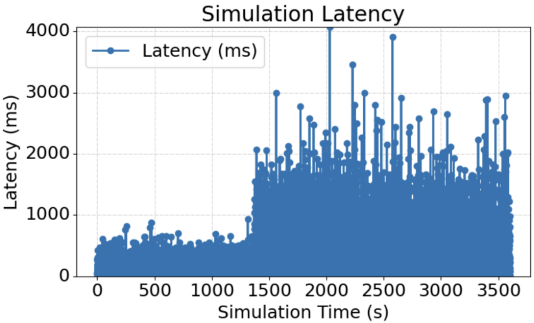
The Performance Cliff

What happens when credits hit zero:

- Simulation shows 5x degradation for CPU-bound queries;
- Production cascading effects often worse.

Takeaway

This is why "database is up" does not mean database is working.



Demo 2: Pool Size Optimization

Question:

How big should my connection pool be?

Demo 2: Testing Pool Sizes

Testing common assumptions:

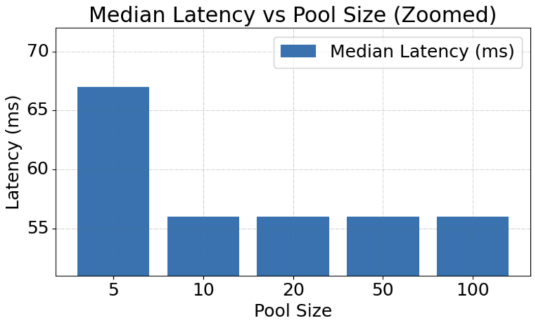
Pool Size	Common Industry Rationale
5	"Too small?"
10	"Rule of thumb: use 10"
20	"2x CPU cores (B2ms has 2 vCPU)"
50, 100	"Just set it high to be safe"

Parameters: Session pooling configuration, 50 requests/sec continuous steady-state load.
Objective: Isolate the minimum pool footprint within 5% of optimal latency.

Demo 2: Results

Key findings:

- **Pool Size 5:** ∼67ms (queueing visible)
- **Pool Size 10:** ∼57ms (optimal for steady load)
- **Pool Size 20–100:** ∼56ms (no latency gain, but handles micro-bursts)



Cross-Cloud Applicability

Industry-standard credit model:

- **Azure B-series**, AWS T-series, GCP E2, Oracle burstable
- All use: earn credits -¿ burn when bursting -¿ throttle when depleted

Minor differences: Initial credits, unlimited mode, baseline %

Same approach works across all cloud providers.

When Simulation Is Worth It

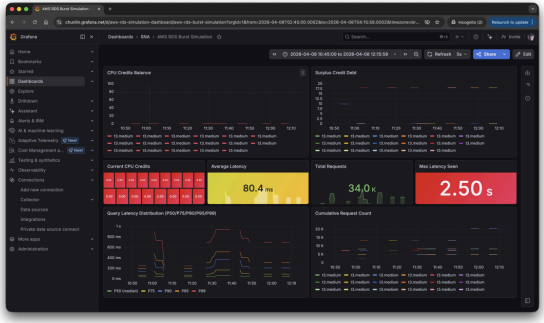
Startup (10 instances): Just guess (config changes are fast)
Enterprise (50+ instances): Simulate (saves $50k/year vs over-provisioning)

This model covers: CPU credits, connection pooling, queueing
Does not model: IOPS throttling, network throttling, storage latency

One More Thing

Observability

Observability: OpenTelemetry Integration



What you're seeing:

- **Credits depleted:** All instances at 0.00 (red boxes)
- **Performance cliff:** Max latency exploded to 2.5 seconds
- **The problem made visible:** Average 80ms looks fine, but p99 catastrophic

SNA exports OpenTelemetry metrics for real-time monitoring

From Guessing to Knowing

Before Simulation	After Simulation
Guess pool size	Calculated optimal size
Test in production	Test in simulation first
Risk of downtime	Identified failure points
Days to tune	1 minute to narrow options
Over-provision 2-5x	Data-driven right-sizing

Thank You

GitHub Repository:
github.com/gcl-team/SNA

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